

People can use the placement of objects to infer communicative goals

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Abstract

Beyond words and gestures, people have a remarkable capacity to communicate indirectly through everyday objects: A hat on a chair can mean it is occupied, rope hanging across an entrance can mean we should not cross, and objects placed in a closed box can imply they are not ours to take. How do people generate and interpret the communicative meaning of objects? We hypothesized that this capacity is supported by social goal inference, where observers recover what social goal explains an object being placed in a particular location. To test this idea, we study a category of common ad-hoc communicative objects where a small cost is used to signal avoidance. Using computational modeling, we first show that goal inference from indirect physical evidence can give rise to the ability to use object placement to communicate. We then show that people from the U.S. and the Tsimane’—a farming-foraging group native to the Bolivian Amazon—can infer the communicative meaning of object placement in the absence of a pre-existing convention, and that people’s inferences are quantitatively predicted by our model. Finally, we show evidence that people can store and retrieve this meaning for use in subsequent encounters, revealing a potential mechanism for how ad-hoc communicative objects become quickly conventionalized. Our model helps shed light on how humans use their ability to interpret other people’s behavior to embed social meaning into the physical world.

Key words: Computational modeling, Social objects, Theory of Mind

1. Introduction

Humans have a remarkable capacity to communicate through objects, even ones we do not usually think of as conveying meaning. A hat on a chair can reveal that the seat is taken; rope surrounding a patch of grass can tell us not to walk through; and, during snowy winters in the northeastern United States, plastic chairs on shoveled parking spots are used to signal that they are not up for grabs. These kinds of everyday objects (Figure 1) do little to physically constrain our actions, yet they affect our behavior because we recognize the meaning they convey. Consistent with this, past empirical research

10 has shown that people spontaneously use objects to communicate (e.g., leaving
11 an open notebook on a library table to mark that the space is occupied; Sommer
12 & Becker, 1969; Becker & Mayo, 1971; Edney & Jordan-Edney, 1974), and
13 detect when an object is communicative (e.g., realizing that the table with a
14 notebook must be taken; Becker, 1973; Shaffer & Sadowski, 1975), with this
15 ability possibly emerging in childhood (Rossano et al., 2015; Zhao et al., 2020).

16 What are the cognitive capacities that support our ability to communicate
17 through objects? One possibility is that communicative objects emerge from a
18 system of simple conventions, where objects and their placement are explicitly
19 associated with different communicative meanings. As children, for instance,
20 most of us likely ignored strap barriers at banks, movie theaters, and DMVs, and
21 their meaning had to be explicitly taught to us. After learning their meaning,
22 we were then able to recall it whenever we encountered them in new locations.

23 While conventional knowledge is undoubtedly a major driver for how we
24 learn and use communicative objects, people are also able to generate novel
25 communicative objects that others can readily understand (such as placing an
26 ironing board to mark that someone has reserved a parking spot; Figure 1i).
27 What computations underlie this capacity? And how does the communicative
28 meaning of novel objects become conventionalized?

29 Here we hypothesized that the capacity to embed and infer communicative
30 meaning from novel objects emerges from our ability to reason about the men-
31 tal states behind other people’s behavior—our *Theory of Mind* (ToM; Wellman,
32 2014; Gopnik et al., 1997). The central idea in our proposal is that, if people can
33 infer other agents’ mental states based on how they manipulated an object (via
34 Theory of Mind), then people can also strategically manipulate objects with
35 the purpose of eliciting mental-state inferences in agents who encounter these
36 objects. Through this method, people can intentionally manipulate their envi-
37 ronment with the goal of communicating their desires to people who navigate
38 the environment when the communicator is absent. We propose that this type
39 of reasoning might support the creation of novel ad-hoc communicative objects,
40 which can then quickly become conventionalized and widespread, supported via
41 memory and recognition.

42 To explore this idea, this paper focuses on a family of objects like those
43 shown in Figure 1. These objects are often not intrinsically communicative:
44 Hats, chairs, and rope are not purposefully designed for communication, but
45 they can nonetheless convey a message when placed in certain locations (Figures
46 1a-c). Moreover, despite their varied use, these objects all communicate some

47 kind of restriction (e.g., “do not use” or “do not cross”). Critically, however, this
48 restriction is not imposed purely through a physical constraint: The cost that
49 these objects impose on agents is low enough that it could be easily ignored (e.g.,
50 walking over the “barriers” in Figures 1b and 1k is trivial). Intuitively, these
51 objects instead work because people realize that the object was intentionally
52 placed with the purpose to communicate. Because of this common structure, we
53 will refer to these objects as *low-cost communicative blockers* (LCCBs). While
54 these objects do not capture the full scope of everyday communicative objects,
55 we believe their use is a fruitful case study for understanding our proposal. We
56 return our focus to communicative objects more broadly in the Discussion.

57 To illustrate the logic of our proposal, imagine trying to find the exit of an
58 unfamiliar building. As you walk down a hallway, you find two doors, side by
59 side. Suppose, however, that one of the doors has a broom positioned diagonally
60 across it. Naturally, it is easy to recognize that (1) someone intentionally placed
61 the broom there and that (2) it creates a small inconvenience for people wanting
62 to walk through the door. When considering why someone would choose to use
63 a broom to block a door, one possibility is that they wanted to prevent people
64 from walking through. But if that were the case, why not put more effort
65 into blocking the door, given how easy it is to move the broom out of the
66 way? Intuitively, this is because their goal was not to create an insurmountable
67 physical constraint—which would require more effort to achieve—but rather to
68 prompt you to infer that they do not want you to walk through.

69 This proposal assumes that people can detect intentional arrangements of
70 objects (e.g., a broom placed diagonally across a door was likely placed inten-
71 tionally), infer what an agent did (e.g., an agent must have taken the broom and
72 placed it there), and determine how much effort it required from the agent and
73 how much it affects us (e.g., how hard was it to place the broom and what effects
74 does this have on my potential plans?). Consistent with this, past research has
75 shown that people have a rich understanding of what physical environments re-
76 veal about people (Gosling et al., 2002; Hurwitz et al., 2019). Moreover, people
77 can infer others’ actions from indirect physical evidence of their presence (Lopez-
78 Brau et al., 2022), and estimate the effort involved in moving and manipulating
79 objects (Yildirim et al., 2019). These capacities also emerge early in develop-
80 ment, with children drawing surprisingly rich inferences from physical evidence,
81 ranging from inferences about what actions an agent took (Jacobs et al., 2021)
82 and what they knew (Pelz et al., 2020) to inferences about even richer social in-
83 formation, such as whether two people transmitted ideas (Pesowski et al., 2020)

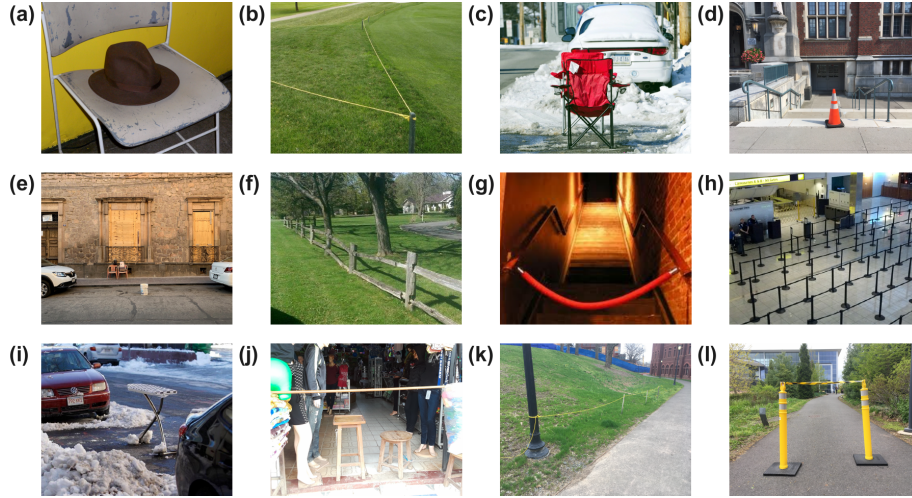


Figure 1: Real-world examples of people communicating through objects. (a) A hat on a chair indicating that someone intends to return. (b) Rope a few inches above the grass so that people know not to walk through. (c) Chairs along the side of the street in South Boston to reveal that someone shoveled and claimed this parking spot. (d) A traffic cone in front of some stairs signaling limited access. (e) A bucket along the side of the street in central Mexico indicating that the parking spot is reserved. (f) An easy-to-cross fence marking a property limit. (g) A stanchion across a stairwell revealing access may be restricted to certain individuals. (h) Belt barriers at the airport telling passengers that they should form a line (and where). (i) An ironing board along the side of the street indicating that the parking spot is taken. (j) A wooden pole and two small benches in a store in Bolivia indicating that the owner is not available. (k) A small rope along a sidewalk asking people not to walk near a construction site. (l) A pair of traffic posts deterring people from using this walkway.

84 and have shared interests (Pesowski et al., 2021).

85 Critically, for communicative objects to have their intended effect, the abil-
86 ity to reason about them is not enough: people must also be motivated to
87 behave cooperatively. If this were not the case, people would ignore low-cost
88 communicative blockers (LCCBs; since the cost they impose is negligible), and
89 communicators would favor creating insurmountable physical constraints rather
90 than communicative signals. While there are undoubtedly cases where people
91 ignore LCCBs, and where people build physical barriers because they do not
92 expect cooperativeness, the pervasive use of these objects suggests that there
93 are many cases where people expect strangers to cooperate by default. This is
94 consistent with evidence that even young children will spontaneously cooperate
95 with strangers (Warneken & Tomasello, 2006) and that adults have a default
96 propensity to cooperate (Rand, 2016).

97 While all this past work establishes the cognitive pre-requisites that our pro-

positional builds on, to our knowledge, no work has yet explored specifically whether these capacities underlie the ability to use communicative objects (Figure 1).

1.1. Paper overview

Our paper has three goals. Our first goal is to test whether our theoretical proposal can, in principle, explain the logic of low-cost communicative blockers (LCCBs), where agents share mental states by using objects to impose a minimal cost on observers. To achieve this, we present a model that explicitly formalizes our proposal in computational terms, and we explore its behavior in synthetic simulations (Sections 2-3) to test whether it can produce patterns that resemble how people use communicative objects. Our computational model focuses on the inferences that we make once an object placement is detected as intentional. We return to the question of how to detect intentional placements in the Discussion.

Having found theoretical support for our proposal, our second goal is to test whether the fine-grained quantitative predictions of our account match human judgments when reasoning about novel low-cost communicative blockers. That is, we use our model to generate exact numerical predictions about the strength of inferences that people should make in different situations, and we compare them to human judgments (Experiment 1).

Finally, our third goal is to test whether the mechanisms we propose play a role when conventional knowledge (i.e., object-meaning mappings that are in common ground for a social group) is unavailable. That is, our goal is not to argue against the critical role of convention, but rather to ask what types of inferences people engage in when they face an object that has no conventional meaning attached to it (Experiments 2-3), and to explore how these inferences become conventionalized (Experiment 4).

While our account proposes mechanisms that support both the creation and understanding of communicative objects, these two behaviors are asymmetrical in two ways. First, as we show in our model below, recognizing the meaning of communicative objects is easier than creating them, requiring one fewer level of recursion. Second, for communicative objects to become ubiquitous, the ability to infer their meaning must be widespread, while the ability to invent them can be restricted to a few individuals. Thus, after confirming the computational plausibility of our account (Sections 2-3), our behavioral studies (Experiments 1-4) focus on people’s ability to infer the communicative meaning of low-cost communicative blockers, rather than on how they are created. We return to this asymmetry in cognitive demand in the Discussion.

134 2. Computational Framework

135 For simplicity and clarity, we describe our model in the context of a simple
136 event similar to the ones we use in Experiments 2-4. Here, an agent (the *decider*)
137 encounters two doors—door *A* and door *B*—and must decide which one to walk
138 through. Before they do, another agent (the *enforcer*), who wants to influence
139 the decider’s choice, has the opportunity to place objects in front of either door,
140 including stacking multiple objects to create a physical constraint. To illustrate,
141 Figure 2a shows a situation where the enforcer has access to four boulders that
142 can be positioned in front of either door. Figures 2d-h show five possible changes
143 that the enforcer could implement (among many others). Our model therefore
144 consists of (i) an enforcer that moves objects in a scene with the goal of affecting
145 a decider’s behavior, and (ii) a decider that determines what to do in a scene
146 by thinking about the costs that the objects in the scene impose. Critically, we
147 assume that the enforcer and decider are never in the scene at the same time,
148 such that the decider only has access to the physical layout of the scene.

149 In this setup, the enforcer can always pursue a simple non-communicative
150 strategy: stack enough objects in front of one of the doors to the point that
151 walking through it is so much work that the decider will prefer to avoid it (e.g.,
152 Figure 2f). However, stacking objects in front of a door is also costly for the
153 enforcer. This creates a preference for more efficient strategies, where agents
154 might exploit their Theory of Mind to use objects in a communicative manner.

155 Under our proposal, people use objects to share mental states by reasoning
156 about the costs incurred by the enforcer (how costly is it for the enforcer to block
157 paths?) and the decider (what costs does this impose on the decider?). Because
158 past work has already studied how people reconstruct behavior from physical
159 displays and estimate the underlying costs (Yildirim et al., 2019; Lopez-Brau
160 et al., 2022; Pesowski et al., 2020), our model takes this capacity for granted
161 and focuses on the inferences that people make given access to these costs.

162 To make cost-based inferences, our framework instantiates Theory of Mind
163 (ToM) as a form of simple recursive social reasoning, similar to models developed
164 to understand pedagogical demonstrations (Ho et al., 2016; Shafto et al., 2014),
165 pragmatics (Frank & Goodman, 2012; Goodman & Frank, 2016), and mental-
166 state inferences (Ullman et al., 2009), and similar to the logic behind k -level
167 ToM models (where k is a variable indicating the recursion depth within ToM;
168 Devaine et al., 2014). At its core, our model is structured around an assumption
169 that agents act to maximize their subjective utilities—the difference between the

costs that they incur and the rewards that they obtain. This assumption is at the heart of human mental-state inferences in adults (Jern et al., 2017; Baker et al., 2017; Jara-Ettinger et al., 2020) and emerges early in development (Liu et al., 2017; Jara-Ettinger et al., 2016; Gergely & Csibra, 2003; Lucas et al., 2014).

Formally, let S be the space of all possible scenes, where each scene $s \in S$ represents an observable arrangement of objects (e.g., see Figures 2a,d-h for six possible scenes in the boulder example). Each agent in this context is defined by two main components. The first is a cost function that captures how agents interact with the environment. For the enforcer, their cost function C_E represents the cost of moving objects, such that $C_E(s_0, s)$ is the cost of transforming an initial scene s_0 into a final scene s (e.g., the cost of changing the scene so that an object in a corner is now in front of a door). For the decider, their cost function C_D represents the cost of navigating the environment, such that $C_D(a, s)$ is the cost of taking action a in scene s (e.g., the cost of walking through a door with an object in the way).

The second main component is a reward function that captures each agent’s desires. The enforcer’s reward function R_E represents their desire to affect the decider’s behavior. That is, $R_E(a)$ is the reward the enforcer obtains when the decider takes action a . The decider’s reward function R_D represents their own personal desires: $R_D(a)$ is the decider’s personal reward when choosing action a (e.g., the decider’s reward when they choose door A in Figure 2).

Our computational framework uses these cost and reward functions to build a model of recursive social reasoning, where the enforcer decides how to move objects by thinking about what action they hope the decider will take, and the decider decides what action to take by inferring what the enforcer wants, based on how they manipulated the objects in the scene. Below, we present the logic of our model, starting with the grounding level of the recursive structure. A more detailed presentation of our model can be found in SM.

2.1. Non-mentalistic decider

The lowest level of our model consists of a non-mentalistic decider D_0 that represents an agent lacking any awareness that objects in a scene may have been intentionally manipulated by another agent. This decider therefore chooses what to do based on the physical properties of the scene alone. Given a scene s , the non-mentalistic decider’s utility for taking action a is given by:

$$U_{D_0}(a; s) = R_{D_0}(a) - C_{D_0}(a, s), \quad (1)$$

where $R_{D_0}(a)$ is the reward that the decider obtains from taking action a and $C_{D_0}(a, s)$ is the cost they incur from taking that action in scene s .

We transform this utility function into a probability distribution over actions by applying the *softmax function*:

$$p_{D_0}(a|s) \propto \exp(U_{D_0}(a; s)/\tau). \quad (2)$$

The softmax function is a standard method for transforming utility functions into probability distributions, guided by a temperature parameter $\tau \in (0, \infty)$. When τ is low, the decider consistently chooses the actions that maximize the utility function (converging towards optimal behavior as $\tau \rightarrow 0$). When τ is high, the decider’s behavior becomes noisier, and the agent is more likely to select actions that are not necessarily the best ones (converging towards random behavior as $\tau \rightarrow \infty$).

2.2. Simple enforcer

The next level of our model consists of a simple enforcer E_0 who reasons about the non-mentalistic decider D_0 . That is, this enforcer determines what to do under the assumption that the decider will not realize that the objects contain any social information and will instead see them as nothing more than physical obstacles.

Formally, suppose the world is in some initial state s_0 and the enforcer wants the decider to take action a (e.g., the initial scene might be Figure 2a, and the enforcer wants the decider to choose door A). To do this, the simple enforcer considers different possible scenes s (e.g., Figures 2d-h) and evaluates them through the utility function:

$$U_{E_0}(s; a, s_0) = \underbrace{R_{E_0}(a)p_{D_0}(a|s)}_{\substack{\text{Expected reward} \\ \text{when decider takes} \\ \text{action } a \text{ in scene } s}} - \overbrace{C_{E_0}(s_0, s)}^{\substack{\text{Cost of transforming} \\ \text{scene } s_0 \text{ into scene } s}}. \quad (3)$$

Here, the first term ($R_{E_0}(a)p_{D_0}(a|s)$) is the enforcer’s expected reward (i.e., the reward they obtain when the decider takes action a , weighted by the probability that the decider takes this action). This enforcer’s ability to predict how the decider will act, $p_{D_0}(a|s)$, is computed using the non-mentalistic decider model

(Equation 2, Section 2.1). This term is then balanced against the cost $C_{E_0}(s_0, s)$ that the enforcer incurs in transforming scene s_0 into scene s . Combined, the first term leads the enforcer to prefer scenes where the decider is more likely to take the desired action, and the second term leads the enforcer to favor minimal scene changes over drastic ones.

2.3. Mentalistic decider

Having defined the simple enforcer E_0 , we can now specify a mentalistic decider D_1 that reasons about this enforcer’s choices. That is, this decider infers why the enforcer decided to modify the scene, and takes this into account when deciding what to do.

Formally, the mentalistic decider assigns a utility to each action via

$$U_{D_1}(a; s_0, s, \phi) = \underbrace{R_{D_1}(a) - C_{D_1}(a, s)}_{\text{Decider's egocentric costs and rewards}} + \overbrace{\phi < \ell(a|s_0, s) >}^{\text{Decider's allocentric preferences}}. \quad (4)$$

The first two terms ($R_{D_1}(a)$ and $C_{D_1}(a, s)$) capture the decider’s egocentric rewards and costs for taking action a in scene s , respectively (identical to the utility function for the non-mentalistic decider; Equation 1, Section 2.1).

The final term, $\phi < \ell(a|s_0, s) >$, represents the decider’s utility for acting in accordance with the enforcer’s preferences, also known as the decider’s “adopted utility” (Powell, 2022). Here, $< \ell(a|s_0, s) >$ is the decider’s belief that the enforcer wants them to take action a , based on the change from scene s_0 to scene s . This term is then weighted by a real-valued cooperation parameter ϕ that captures the decider’s motivation to pursue, act against, or ignore the enforcer’s preferences. When ϕ is positive, the decider is motivated to act in a way that is consistent with the enforcer’s preferences. Conversely, when ϕ is negative, the decider is antagonistic and prefers to act against the enforcer’s preferences. Finally, when $\phi = 0$, the decider acts egocentrically (becoming the same model as the non-mentalistic decider), and treats objects as physical constraints, ignoring why the enforcer might have positioned them there.

Critically, the decider does not know a priori what the enforcer wants them to do (i.e., the decider does not have direct access to $\ell(a|s_0, s)$). This term is therefore inferred by considering the enforcer’s possible reward functions:

$$< \ell(a|s_0, s) > = \int_{R_{E_0} \in R} \mathbb{1}_{\arg \max R_{E_0}}(a) p(R_{E_0} | s_0, s). \quad (5)$$

260 This equation adds up the probability of every possible reward function $R_{E_0} \in R$
 261 where a is the preferred action. For each of these reward functions, its probabil-
 262 ity, $p(R_{E_0}|s_0, s)$, is inferred by reasoning about the enforcer’s choice to change
 263 scene s_0 into scene s :

$$p(R_{E_0}|s_0, s) \propto p_{E_0}(s|R_{E_0}, s_0)p(R_{E_0}), \quad (6)$$

264 with the likelihood $p_{E_0}(s|R_{E_0}, s_0)$ computed using the simple enforcer model
 265 (i.e., it is given by the softmax of Equation 3, Section 2.2).

266 Note that this formulation assumes that the mentalistic decider knows both
 267 the scene’s initial and final states (s_0 and s). This allows deciders to infer
 268 the enforcer’s rewards by reasoning about the costs that were introduced. In
 269 more realistic situations, it is more likely that deciders have a prior distribution
 270 over scenes ($p(s_0)$) rather than perfect knowledge about the initial scene s_0 .
 271 Returning to the example in the introduction, for instance, when encountering
 272 a broom placed across an entrance, a decider may not know where the broom
 273 was situated before an enforcer placed it across the door, but they may believe
 274 it was more likely that it was positioned elsewhere. Modeling prior expectations
 275 about scene distributions is beyond the scope of our model, but we return to
 276 the implications of this assumption in the Discussion.

277 2.4. Complex enforcer

278 Finally, we can define a complex enforcer E_1 that modifies scenes by thinking
 279 about a mentalistic decider D_1 . This model is identical to the simple enforcer
 280 (Section 2.2), with the only difference that it predicts the decider’s behavior us-
 281 ing the mentalistic decider model (Section 2.3), rather than the non-mentalistic
 282 one. That is, the term $p_{D_0}(a|s)$ from Equation 1 is now replaced by $p_{D_1}(a|s, \phi)$
 283 (i.e., the softmax of the utility function in Equation 4). This enforcer can there-
 284 fore manipulate scenes under the assumption that the decider will attempt to
 285 decode their preferences.

286 2.5. Model implementation details

287 The computational framework specified above captures the proposal that
 288 people extract costs from physical scenes (i.e., what is the cost of taking different
 289 actions, and what costs did another agent incur in positioning the objects),
 290 and use them to make mental-state inferences. Because past work has already
 291 studied how people might infer costs from physical scenes (Yildirim et al., 2019),

our interest is in testing how cost manipulations shape communication with objects. Therefore, in our model, we directly provide the costs associated with each scene change, which enables us to focus on the contribution of cost-based reasoning.

In principle, the parameters in our model can all be real-valued. For simplicity, we bounded costs and rewards to integers in the range 0 to 9. This enables us to easily interpret the range with 0 being null costs and rewards, and 9 being the highest possible costs or rewards that agents can have. We next set the cooperation parameter ϕ to take on integers between -25 and 25, which allows the model to consider extreme cooperative and adversarial cases (see Oey et al., 2022, for related work on adversarial mental-state reasoning). Our model code is available online at <https://osf.io/57n4g>.

3. Model Analysis

Our first goal is to use our computational model to test whether our proposal can capture the emergence and use of low-cost communicative blockers (LCCBs). If our model failed to replicate this phenomena, this would imply that our account is incorrect. Specifically, our analyses consist of a set of simulations that test whether the enforcer and decider in our model can reproduce our target phenomena—creating and understanding LCCBs (inspired by those in Figure 1).

To explore our model dynamics, we focused on the same simple domain with two doors—door *A* and door *B*—and an enforcer that wants the decider to choose door *A* (Figure 2). We assume that the initial scene s_0 has a set of objects between the two doors, such that the objects do not initially block either door, and placing an object in front of either door is equally costly (Figure 2a). For simplicity, we also assume that the cost the enforcer incurs in placing an object in front of a door is the same as the cost that the decider incurs when moving that object out of the way. To analyze the core dynamics of the model, we simulated a situation where the enforcer was maximally motivated to affect the decider’s behavior (setting their reward to 9), and where the decider was also very cooperative (setting $\phi = 10$). We further set the softmax parameter to a minimum in order to remove any noise in the inferences (setting $\tau = 0.1$). We then tested our model’s performance by varying the decider’s relative preference for different options.

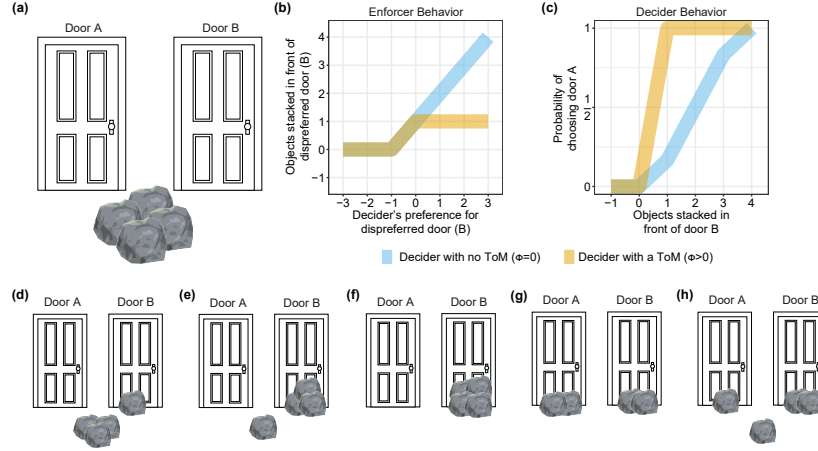


Figure 2: Example event used to illustrate model performance. The environment consists of two doors and a stack of boulders between them. The enforcer’s goal is to reposition the boulders to get the decider to choose door A. (a) Initial scene state. (b) Enforcer behavior. The x -axis shows the decider’s preference for door B (negative values indicating a preference for door A) and the y -axis shows the number of boulders the enforcer stacks in front of door B (negative values indicating stacking objects on door A). The simple enforcer (blue line) builds the smallest possible physical barrier that will dissuade the decider. The complex enforcer (yellow line) places a single boulder in front of door B, even when the decider has a strong preference for going through it. (c) Deciders with a preference for door B reacting to boulders placed in front of that door. The non-mentalistic decider (blue line) slowly becomes more likely to choose door A as a function of how many boulders are blocking door B. The mentalistic decider (yellow line) recognizes the meaning of a single boulder and adjusts their behavior, immediately forgoing their preferred door B and choosing door A instead. (d-h) Visualization of some of the different scenes the enforcer could produce.

Figure 2a shows a visual depiction of the initial state, and Figures 2d-h shows five possible scene transformations that the enforcer could produce (stacking one, three, or four objects in front of door B alone, or also stacking any number of objects in front of door A). To understand our model behavior, we began by contrasting the simple and complex enforcers. Figure 2b shows how many objects each enforcer chooses to stack in front of door B (the door they hope the decider will avoid) as a function of the decider’s preference for this door. When the decider already prefers door A (negative decider preference along the x -axis in Figure 2b), neither enforcer moves any objects. This reflects the enforcers’ confidence that the decider will take door A, making any involvement unnecessary.

When the decider prefers door B (positive decider preference along the x -axis in Figure 2b), the enforcers begin to place objects in front of the door, producing two different types of behavior. The simple enforcer expects the

340 decider to choose a door based only on their egocentric costs (how difficult is
341 it to walk through each door?) and rewards (how much does the decider want
342 to walk through each door?). Consequently, this enforcer stacks the minimum
343 number of objects necessary to push the decider’s choice towards door A . This
344 is captured in Figure 2b, where the blue line shows how the simple enforcer
345 stacks more objects as the decider’s preference becomes stronger. This behavior
346 reflects a non-communicative barrier-building strategy, where the enforcer is
347 attempting to make it just hard enough for the decider to cross through door
348 B , with the hope that this added cost will shift their preference towards door
349 A .

350 In contrast to the barrier-building strategy from the simple enforcer, the
351 complex enforcer places a single object in front of door B (as in Figure 2d),
352 even when the decider really prefers that door (yellow line in Figure 2b). We
353 interpret this as the kind of communicative strategy that we aim to explain
354 (reminiscent of Figure 1): The strategy succeeds not because it imposes a high
355 cost on deciders, but because it efficiently reveals the enforcer’s mental states.
356 In these cases, the enforcer knows that the decider’s egocentric utilities will
357 favor door B , because the single object imposes a negligible cost. The enforcer
358 nonetheless chooses to place a single object in front of door B because they
359 believe that the decider will infer that they are supposed to take door A instead.

360 Returning to our motivating examples (Figure 1), this behavior resembles
361 actions like placing a plastic chair to mark that a parking spot is taken. Here, a
362 plastic chair does little to prevent someone from using the parking spot: moving
363 the chair out of the way is easy, and the cost is probably insufficient to overcome
364 a driver’s desire to find a parking spot. However, the object is effective because
365 it reveals that whoever placed the chair is requesting that their parking spot be
366 respected.

367 Figure 2c shows the behavior of our decider model. The non-mentalistic
368 decider responds to the physical costs alone, becoming more likely to abandon
369 their preferred door as a function of how many objects are blocking it. This is
370 visualized by the blue line in Figure 2c, which shows a continuous preference
371 change as a function of the number of objects blocking their preferred door. By
372 contrast, the mentalistic decider shows a sharp discontinuity: A single object in
373 front of their preferred door is enough for them to understand that they should
374 avoid that door. This is visualized by the yellow line in Figure 2c, where the
375 decider shows a rapid change in strategy as soon as a single object is in front
376 of their preferred door. Together, these results show how our model gives rise

377 to enforcers who use objects in a communicative manner and deciders who can
378 infer the communicative meaning of these objects.

379 4. Experiment 1: Quantitative model evaluation

380 Having established that our account can replicate the qualitative use and
381 recognition of low-cost communicative blockers (LCCBs), we next test whether
382 our model’s exact inferences match human intuitions. That is, our model pre-
383 dicts quantitative patterns about how strong people’s intuitions should be in
384 different displays. If participants can interpret LCCBs, but do so in a different
385 way than our model does, the resulting large discrepancies between our model
386 inferences and participant judgments would falsify our account.

387 In Experiment 1, participants saw a two-dimensional gridworld of a fruit
388 farm with an entrance, pomegranate groves, pear groves, and a set of boulders
389 placed by a farmer to protect their pomegranates from nearby hikers (farm-
390 ers corresponding to enforcers and hikers to deciders from our Computational
391 Framework).

392 We tested participants in two conditions (Figure 3). In the *non-mentalistic*
393 condition, hikers believe that the boulders are natural constraints, devoid of
394 social meaning, and farmers plan for how many boulders to place accordingly.
395 We therefore expect participants to infer that, the more boulders the farmer
396 places, the more she believes that hikers want to take the pomegranates. We
397 model this condition using the non-mentalistic decider model (Section 2.1) and
398 the simple enforcer model (Section 2.2).

399 In our second condition, the *mentalistic* condition, hikers will always know
400 that the boulders were placed by a farmer, and use the costs imposed by these
401 objects to infer the farmer’s preferences. In this condition, a single boulder does
402 not necessarily imply an expectation that hikers do not like pomegranates (as
403 would be implied in the *non-mentalistic* condition). Instead, a single boulder
404 might reveal that the farmer expects hikers to infer that they should stay away
405 and act accordingly. By contrast, if the farmer placed multiple boulders, this
406 would reveal that she expects hikers to prefer pomegranates and be uncoop-
407 erative (otherwise, a single communicative boulder would have sufficed). We
408 model this condition using the mentalistic decider model (Section 2.3) and the
409 complex enforcer model (Section 2.4).

410 All studies were approved by Yale’s IRB (protocols “Culture and Cognition”
411 #2000022403 and “Online reasoning” #2000020357). Data collection was ob-

412 tained in the following experiment order: 3a (meaning inference), 3b, 2, 4, 1,
413 1 replication, 3a (unusualness ratings), and 3c. Our experimental procedure,
414 stimuli, data, analyses, and pre-registrations (for Experiments 1 replication, 2,
415 3a, and 3c) are available at <https://osf.io/57n4g>. This manuscript includes
416 all experiments, manipulations, and measures in this line of research.

417 4.1. *Participants*

418 80 U.S. participants (as determined by their IP address) were recruited using
419 Amazon Mechanical Turk ($n = 40$ per condition; $M = 34.81$ years, $SD = 10.31$
420 years).

421 4.2. *Stimuli*

422 Stimuli consisted of 27 10-by-10 gridworlds, with two fruit groves (pears and
423 pomegranates), a hiker, and a set of boulders (see Figures 5a-c for examples).
424 The stimuli were designed by parametrically varying two factors: the distance
425 between the hiker and the groves (i.e., the natural cost of the environment; 5,
426 7, and 9 squares away) and the number of boulders blocking the pomegranates
427 (i.e., the artificial cost introduced by the farmer; 1, 2, or 3 boulders). The
428 hiker's starting position was randomly selected to be at one of the four corners,
429 and the fruit groves were randomly placed on the two adjacent corners relative
430 to the hiker.

431 4.3. *Procedure*

432 Participants read a brief cover story explaining that they would see hikers in
433 different farmlands with pear and pomegranate groves (see Figure 3 for paradigm
434 schematic). The farmers, who were absent, did not mind hikers taking pears
435 but they wanted to protect their pomegranates. To achieve this, farmers placed
436 boulders in front of their pomegranate groves (see Figures 5a-c for examples).
437 Participants then completed a multiple-choice five-question quiz (see online OSF
438 repository for questions) to ensure they understood the task. Participants that
439 answered at least one question wrong were sent to the beginning of the cover
440 story to try again. Participants that failed the questionnaire twice were not
441 permitted to participate in the study.

442 Participants in the *non-mentalistic* condition were told that hikers thought
443 the boulders were natural constraints, and that farmers planned how many
444 boulders to place accordingly. That is, the farmer expected hikers to realize
445 that the boulders make it harder to reach a fruit grove, but assume that this

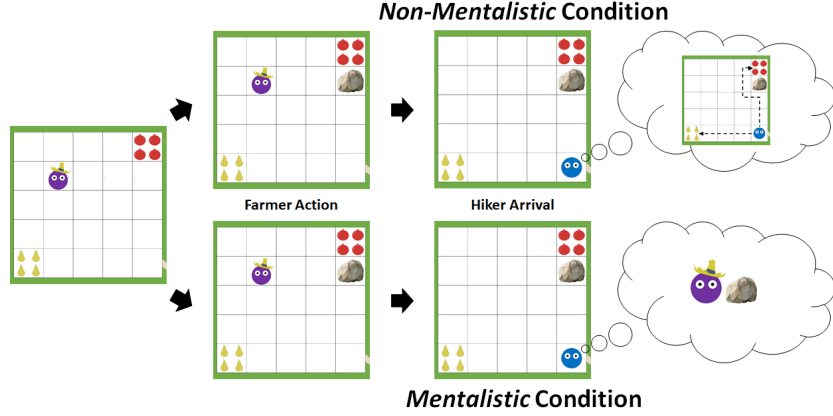


Figure 3: Visual schematic of Experiment 1 cover story. Participants learned that a farmer (purple agent) wanted to protect their pomegranates and placed boulders to block the way before leaving. After leaving, a hiker would arrive and decide which fruit to take. In the *non-mentalistic* condition, the hikers treat the boulders as natural constraints, and they therefore decide what to do without thinking about the farmer. In the *mentalistic* condition, the hikers know that a farmer must have placed the boulders, and use this to infer what to do.

was simply a feature of the terrain, rather than an intentional design. In each trial, participants saw an arrangement of boulders and they were asked how much the farmer expected hikers to like pomegranates (“How much does this farmer think that hikers like pomegranates?”), using continuous sliders ranging from “not at all” to “very much”).

Participants in the *mentalistic* condition were told that hikers would always know that a farmer placed the boulders intentionally, and that farmers planned how many boulders to place accordingly. That is, the farmer expected hikers to know that the boulders make it harder to reach a fruit grove, and that these boulders were placed intentionally by someone. In each trial, participants saw an arrangement of boulders and they were asked how much the farmer expected the hiker to like pomegranates. In addition, because the complex enforcer and mentalistic decider include a cooperation parameter ϕ (i.e., the adopted utility weight; Powell, 2022), participants were also asked whether the farmer expected the hiker to be cooperative (“How cooperative does this farmer think hikers are?”), using a continuous slider ranging from “not at all” to “very much”).

All participants completed the same 27 trials (trial order randomized across participants), where we varied both the initial cost of obtaining each type of fruit (by manipulating the initial distance from the hiker) and the number of

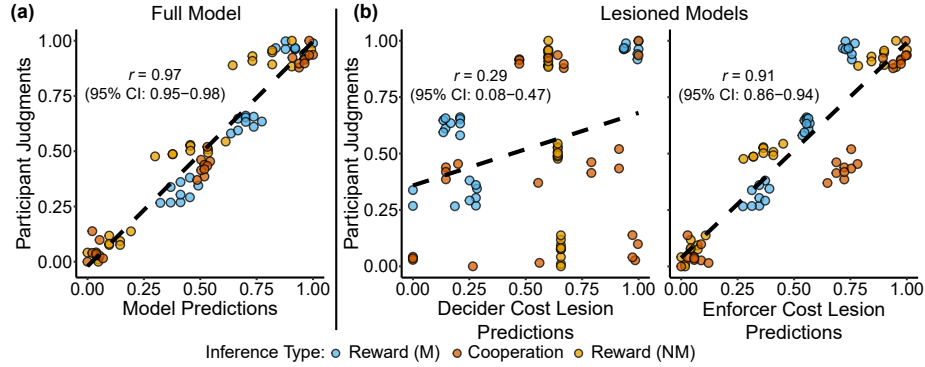


Figure 4: Experiment 1 results. Each point represents a judgment, with model predictions on the x -axis and participant judgments on the y -axis. Participants in the *non-mentalistic* condition (NM) inferred the hiker’s expected preferences, and participants in the *mentalistic* condition (M) additionally inferred the hiker’s expected cooperativeness (ϕ). (a) Correlation between our full model and participant judgments. (b) Correlation between lesioned models and participant judgments. Model lesions include removing the influence of cost from the decider (left) and removing the influence of cost from the enforcer (right).

465 boulders that the farmer added (ranging from 1 to 3; see Stimuli).

466 4.4. Model Predictions

467 Our model’s parameters were set prior to data collection (and reflected in
 468 the pre-registration of the Experiment 1 replication; see Section 2.5 and SM
 469 for details). For each dependent variable in our task we computed our model’s
 470 posterior predictive distribution, and used the expected value as the final model
 471 prediction.

472 4.5. Results

473 Our model and participant judgments showed an overall correlation of $r =$
 474 0.97 ($CI_{95\%}: 0.95 - 0.98$; Figure 4a). A pre-registered replication of this study
 475 produced identical results ($r = 0.98$; $CI_{95\%}: 0.96 - 0.98$; see SM for details).
 476 The fact that our model captures the fine-grained structure of people’s inferences
 477 suggests that their inferences resembled the ones obtained by reasoning about
 478 the farmer’s desires via recursive social reasoning.

479 Figure 5 shows three example trials that highlight the inferences that our
 480 model and participants made. In Figures 5a-c, the hiker’s distance to pomegran-
 481 ates and pears is matched (making the initial cost identical) and the number of
 482 boulders in front of the pomegranates varies from 1 to 3. In the *non-mentalistic*
 483 condition, the number of boulders should reveal how much the farmer thinks

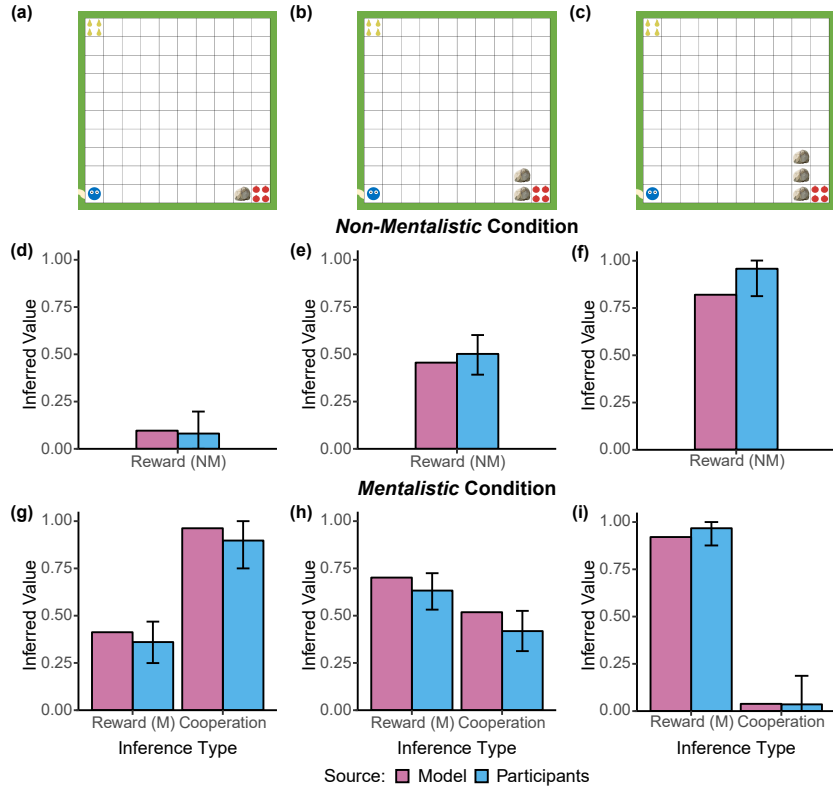


Figure 5: (a-c) Example stimuli from Experiment 1. In these examples, both fruit groves were equally far and only varied on the number of boulders a farmer placed. (d-f) Model predictions and participant judgments from the *non-mentalistic* condition in purple and blue, respectively. (g-i) Model predictions and participant judgments from the *mentalistic* condition in purple and blue, respectively. Inference type is along the *x*-axis and the inferred value is along the *y*-axis. Error bars are bootstrapped 95% CIs.

hikers will want to get the pomegranates (because the purpose of the boulders is to introduce a physical cost that outweighs the hikers' desires). Figures 5d-f show this effect in both our model and participant reward inferences (with each plot corresponding to the stimuli directly above it; e.g., panel (d) corresponding to map (a)). As the number of boulders increased, participants and our model inferred a stronger preference for pomegranates.

In the *mentalistic* condition, the boulders not only impose a physical cost, but allow hikers to infer the farmer's preferences. Therefore, a single boulder (Figure 5g) does not necessarily imply that hikers must not like pomegranates that much (as it did in the *non-mentalistic* condition; Figure 5d). Instead, the farmer may have used a single boulder to reveal that they did not want

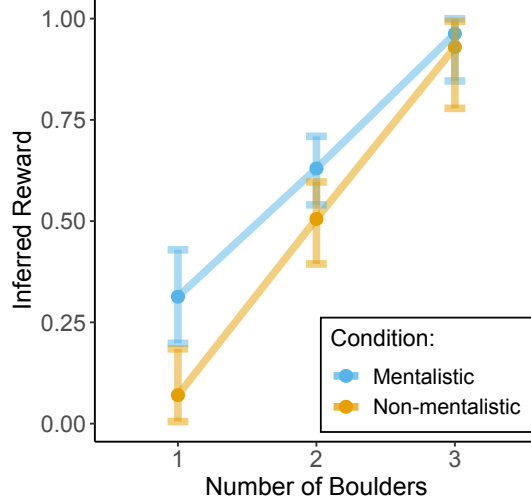


Figure 6: Reward inferences across the *non-mentalist* condition and *mentalist* condition in Experiment 1. The number of boulders placed by the farmer is on the *x*-axis and the reward participants inferred is on the *y*-axis. Error bars are bootstrapped 95% CIs.

hikers to take the pomegranates. Consistent with this, both participants and our model inferred that hikers could have a higher desire for pomegranates (compare Figures 5g and 5d), but were highly cooperative. That is, participants and our model inferred that a single boulder was effective because it revealed the farmer’s preferences to cooperative hikers (despite its cost not being high enough to outweigh their preferences). This reward difference across conditions is further visualized in Figure 6, and was significantly different across conditions ($\Delta R = 0.24$; $p < 0.001$ from a two-tailed *t*-test).

When the number of boulders blocking the pomegranates increases (Figures 5h-i), the farmer’s additional actions (placing more than one boulder) can be explained by inferring that hikers must really want the pomegranates and not be particularly cooperative (given that they will be able to infer that the farmer wants them to stay away). Consistent with this, both our model and participants infer a stronger hiker desire and a lower cooperativeness as the number of boulders increases (see SM for additional results of a linear mixed-effects regression predicting these participant reward inferences as a function of boulder count and condition).

512 4.5.1. Alternative models

513 While our model captured participant inferences with quantitative accuracy,
 514 it is possible that participants reached similar inferences through simpler mech-
 515 anisms. To test this, we considered two alternative models. A first possibility
 516 is that people focus only on an object’s position, without considering the costs
 517 that it might impose on observers. In our experiment, this means that hikers
 518 do not consider the cost of navigating around boulders. We call this model
 519 the *Decider Cost Lesion* as it is similar to our model with the difference that
 520 it does not reason about the cost that objects impose on deciders. Figure 4b
 521 (left) shows how this model was no longer able to explain participant judgments
 522 ($r = 0.29$; $CI_{95\%}$: $0.08 - 0.47$), and was also reliably worse than our main model
 523 ($\Delta r = 0.68$; $CI_{95\%}$: $0.49 - 0.89$). This result confirms that the cost imposed on
 524 deciders is critical for capturing human-like inferences.

525 A second possibility is that people do consider the costs that an object
 526 imposes on their actions (e.g., detecting that an object is making it harder for
 527 them to get a certain fruit), but they do not consider the effort that someone
 528 had to incur in positioning the object. In our experiment, this means that
 529 people do not think about the cost farmers incur when placing boulders. We
 530 call this model the *Enforcer Cost Lesion* as it is similar to our model with the
 531 difference that it does not reason about the cost the enforcer incurs. Figure
 532 4b (right) shows how this lesioned model compares to participant judgments.
 533 Although this model performed worse than our main model ($\Delta r = 0.06$; $CI_{95\%}$:
 534 $0.03 - 0.10$), it was nonetheless able to capture the pattern of inferences about
 535 deciders quite well ($r = 0.91$; $CI_{95\%}$: $0.86 - 0.94$).

536 These results suggest that participant inferences may not depend as heavily
 537 on the cost incurred by the enforcer (i.e., the farmer). We believe this is intuitive
 538 for the situations that we focused on. For instance, when encountering a broom
 539 positioned directly across a door, we intuitively focus on the cost that the broom
 540 imposes on us, rather than thinking about the cost the enforcer incurred. At the
 541 same time, Figure 4b (right) reveals that this lesioned model nonetheless fails
 542 to capture a subset of participant intuitions that our main model was able to
 543 capture. Specifically, this lesioned model over-estimated hikers’ cooperativeness
 544 when compared to humans in trials with two boulders (visualized as a cluster of
 545 orange points that falls most distant from the best-fit line in Figure 4b, right).
 546 This is because, according to this lesioned model, placing three boulders is as
 547 easy as placing two boulders. Therefore, the farmer choosing not to place an

548 extra boulder at no cost would only be reasonable if the hikers were highly
549 cooperative, to the point that placing a single boulder was guaranteed to be
550 as effective as placing more boulders to block the way. Together, this analysis
551 suggests that the cost incurred by enforcers is less critical for capturing how we
552 infer the communicative meaning of an object, but that people are nonetheless
553 sensitive to it, and use it to infer other agents' cooperativeness. Overall, these
554 lesions show how considering the cost that enforcers incur in positioning objects,
555 and how these objects also impose a cost on deciders, are key to explaining how
556 participants reasoned about objects in our experiment.

557 5. Overview of Experiments 2-4

558 Our model analyses and Experiment 1 show that people can derive infer-
559 ences that are quantitatively similar to those from our model. While these
560 results show that people can make these types of social inferences, they do not
561 imply that this is what people do when encountering communicative objects. In
562 Experiments 2-4, our goal is therefore to test an alternative hypothesis: Could
563 simple conventions without inference explain the use of communicative objects
564 in their entirety? That is, conventional knowledge is undoubtedly critical to the
565 everyday use of communicative objects. Our goal is therefore not to question
566 its importance, but to ask what happens when conventional knowledge is un-
567 available, such as when we encounter an unfamiliar object that might have a
568 communicative purpose. In these cases, do people rely on social inferences like
569 the ones we proposed? Or are they unable to make any conclusions given the
570 absence of an explicit convention?

571 Given that people will use conventional knowledge when it is available, our
572 experiments here focus on objects that are not associated with a pre-existing
573 communicative meaning. We begin by testing two predictions. First, if the
574 meaning of communicative objects were based on explicit convention alone,
575 then people should detect an object as communicative only when they have
576 been explicitly taught about its meaning (therefore falsifying our account). By
577 contrast, our account predicts that people should be more likely to associate low-
578 cost novel objects with a communicative purpose, relative to novel objects that
579 impose no cost (as these fail to reveal the mental states of whoever positioned
580 the object). We test this prediction in Experiment 2. Second, if the meaning
581 of objects were driven by explicit pedagogy and convention alone, then people
582 should be unable to infer the meaning of a novel object, even when they know

583 that the object has a communicative purpose (therefore falsifying our account).
 584 By contrast, our account predicts that people should be able to infer the com-
 585 municative meaning of an object when its placement (i.e., the cost it imposes)
 586 reveals the enforcer’s mental states. We test this prediction in Experiment 3.
 587 Finally, if people are engaging in social inferences to infer the meaning of novel
 588 objects, our work brings forth the question of how quickly these meanings might
 589 become conventionalized. In Experiment 4, we test the idea that people might
 590 be able to quickly treat the meaning of a novel object as conventional.

591 6. Experiment 2: Are objects that impose a cost more likely to be 592 communicative?

593 If people’s reasoning about low-cost communicative blockers is driven en-
 594 tirely by explicit object-meaning conventions, then people should report that an
 595 object is communicative only when they have been explicitly taught its mean-
 596 ing. In Experiment 2, we therefore tested whether people believe that objects
 597 that impose a low cost on deciders are more likely to be communicative relative
 598 to objects that do not impose a cost, as our account predicts but the explicit
 599 convention account does not.

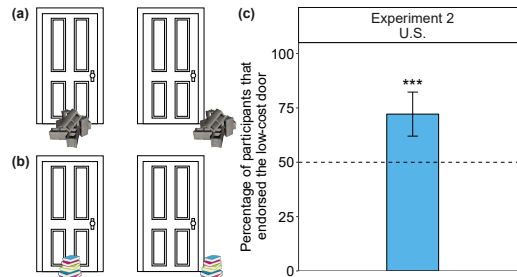


Figure 7: (a-b) Example stimuli from Experiment 2. In both panels, the left side shows a low-cost door and right side shows a no-cost door. (c) Experiment 2 results. The blue bar represents the percentage of participants that associated the low-cost door with having a communicative purpose. Error bars are bootstrapped 95% CIs.

600 6.1. Participants

601 80 U.S. participants (as determined by their IP address) were recruited using
 602 Amazon Mechanical Turk ($M = 37.84$ years, $SD = 12.22$ years). 14 additional
 603 participants were recruited and replaced for failing our inclusion criteria (see
 604 Results).

605 6.2. Stimuli

606 Stimuli consisted of eight images of pairs of doors, with each pair consisting
607 of a “low-cost door” and a “no-cost door” (e.g., Figures 7a-b). Each of the eight
608 pairs was associated with one of eight objects that are not conventionally used
609 to communicate: a plant, a chair, a pile of books, a pile of cinderblocks, some
610 tape, some meter sticks, a hat, and a fishbowl tied to a tack on a door frame (see
611 online OSF repository for the full stimuli set). In the low-cost doors, the object
612 was placed directly in the middle of the doorway (e.g., Figures 7a-b, left), and
613 in the no-cost doors, the object was placed next to the door, not blocking the
614 way (e.g., Figures 7a-b, right). Half of the door pairs were open and the other
615 half were closed.

616 6.3. Procedure

617 Participants were asked to imagine leaving an office and encountering a
618 pair of doors, each with an object nearby. Participants then answered a simple
619 multiple-choice attention-check question (“What objects are in front of the
620 doors?”). Participants that answered incorrectly were sent to the beginning of
621 the cover story and not permitted to access the experiment until they answered
622 correctly.

623 Participants then saw a single trial containing a low-cost door and a no-
624 cost door, both with the same object nearby (see Figures 7a-b for examples;
625 door order randomized across participants). Participants were asked, “Which
626 door do you think someone was trying to tell you something?”, followed by a
627 manipulation check (“Which door requires more work to walk through?”) and
628 an inclusion question (“Do you think you would be able to walk through this
629 door if you wanted to?”). These questions were always presented in the same
630 order (see online OSF repository for the full procedure details).

631 6.4. Results

632 Participants who did not think they could walk through the doors were
633 excluded from the study and replaced (as our interest is in the inferences people
634 make when objects are not seen as insurmountable physical constraints; $n = 14$;
635 14.89% exclusion rate). Of our final sample, 72.50% of participants reported
636 that the low-cost door was more likely to be communicative (CI_{95%}: 62.50% –
637 82.50%; $p < 0.001$ from a two-tailed binomial test; Figure 7c), rather than
638 performing at chance, as expected by the explicit pedagogy account (see SM
639 for a supplemental analysis confirming this result). Our exclusion rate (14.89%)

was lower than recent estimates of attentiveness on Mechanical Turk (estimated to be at approximately 20%; Arechar & Rand, 2021), suggesting that these participants were simply inattentive. However, these participants showed the same qualitative pattern of responses as those included in the task (see SM for details on excluded participants). It is therefore possible that these participants were attentive but did not interpret our inclusion question (“Do you think you would be able to walk through this door if you wanted to?”) as referring to physical plausibility alone, integrating social expectations as well (given the heavy social focus of the task).

7. Experiment 3a: Are low-cost objects interpreted as communicative blockers?

Having found that people are more likely to interpret a low-cost object as communicative, in Experiment 3a, we next test what meaning people are more likely to associate with it. If the object is unfamiliar, the explicit convention account predicts that participants should be at a loss about what it means, given the absence of an established object-meaning mapping. Instead, if people are making inferences about why someone would place the object to impose a cost, they should infer that the object is more likely to mean that they should avoid the door.

7.1. Participants

160 U.S. participants (as determined by their IP address) were recruited via Amazon Mechanical Turk ($n = 80$ per condition; $M = 34.85$ years, $SD = 8.38$ years). 17 additional participants were recruited and replaced for failing our inclusion criteria (see Results).

7.2. Stimuli

Stimuli consisted of 16 images of pairs of doors, with each pair consisting of an empty door and a door with an object nearby (e.g., Figures 8a-b; using the same objects from Experiment 2). In the “low-cost pair”, the object was placed directly in the middle of one of the doorways (e.g., Figure 8a), and in the “no-cost pair”, the object was placed next to one of the doors, not blocking the way (e.g., Figure 8b). Half of these door pairs were open and the other half were closed.

7.3. Procedure

The procedure was similar to Experiment 2. Participants were asked to imagine leaving an office building and finding two identical exits. Participants then saw that one of the doors had an object nearby, and they were told that it was unclear whether someone wanted them to take that door or to avoid it. Participants were then asked a multiple-choice attention-check question: “What is the only difference between the two exits?” Participants next saw a pair of doors (either a low-cost pair for participants randomly assigned to the *low-cost* condition or a no-cost pair for participants randomly assigned to the *no-cost* condition; door order randomized across participants), and were asked: “What do you think someone was trying to tell you about the door with the OBJECT?” (possible responses: “You should walk through the door with the OBJECT” or “You should not walk through the door with the OBJECT”). Participants then responded to the same manipulation-check and inclusion questions from Experiment 2.

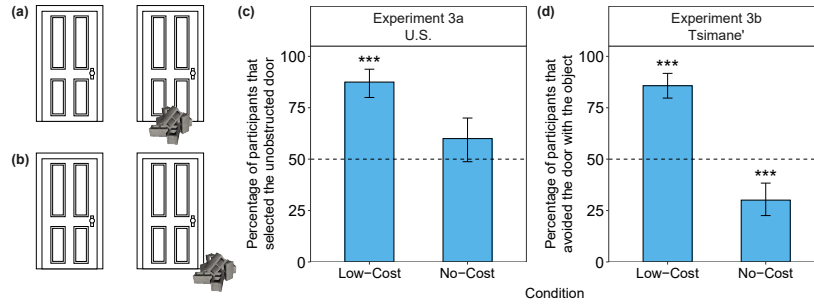


Figure 8: (a) Example stimuli from the *low-cost* condition in Experiment 3a. (b) Example stimuli from the *no-cost* condition in Experiment 3a. (c) Experiment 3a results. The blue bars represent the percentage of U.S. participants that selected the empty door as a function of condition. (d) Experiment 3b results. The blue bars represent the percentage of Tsimane’ participants that inferred that they should not go through the door as a function of door type. Error bars are bootstrapped 95% CIs.

7.4. Results

Participants who said the empty door was harder to walk through were excluded from the study and replaced ($n = 17$; 9.60% exclusion rate). Of our final sample, 87.50% of participants in the *low-cost* condition inferred that they were supposed to avoid the door with the object ($CI_{95\%}$: 80.00% – 93.75%; $p < 0.001$ from a two-tailed binomial test). By contrast, only 60.00% of participants in the *no-cost* condition inferred that this door should be avoided ($CI_{95\%}$:

48.75% – 70.00%; $p = 0.093$ from a two-tailed binomial test¹), a proportion not significantly different from chance. Moreover, the number of participants inferring that they should avoid the door was significantly higher in the *low-cost* condition relative to the *no-cost* condition ($p < 0.001$ by Fisher’s exact test). The fact that participants did not perform at chance in both conditions suggests that people do not rely purely on conventional knowledge. The pattern of results from our excluded participants was qualitatively consistent with that of participants included in the task (see SM for details on excluded participants).

Our experiments so far are consistent with our proposal: participants may be engaging in cost-based reasoning when interpreting the meaning of low-cost communicative blockers. However, there are two alternative explanations that are important to consider. First, is it possible that the position of an object broadly indicates what space it concerns, independent of the cost it imposes? That is, rather than participants reasoning about object costs, they may arrive at similar inferences by seeing any inapposite object in front of a door.

To test this possibility, we conducted a secondary analysis where we separated trial types within our experiment. Our overall experimental design used eight objects in a *low-cost* and a *no-cost* condition. For two of the eight objects, both the *low-cost* and *no-cost* variants had the object in front of the door (Figure 9). If the inapposite account is correct, then participants should infer avoidance in both the *low-cost* and *no-cost* versions for these two objects. In these trials, participants inferred avoidance in the *low-cost* condition significantly above chance (19/20; $p < 0.001$ from a two-tailed binomial test), but were at chance in the *no-cost* condition (14/20; $p = 0.120$ from a two-tailed binomial test). This suggests that having an object in front of a space is insufficient for triggering these inferences.

A second possibility is that people simply believe that more ‘unusual’ arrangements of objects are more likely to signal avoidance. To answer this question, we conducted an additional experiment where we asked participants to rate how unusual each arrangement of objects appeared to them, and we repeated our analyses while controlling for unusualness. In this analysis, we found that the cost that an object imposes continued to significantly predict participant answers, independent of the object’s unusualness (see Section 5 in SM).

¹Our pre-registered analysis proposed to use logistic regressions to study this effect. We instead present binomial tests for clarity, but the results are identical under our pre-registered analysis and can be found in SM.

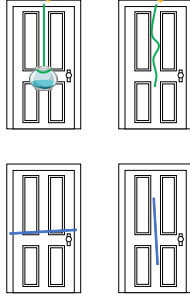


Figure 9: Two pairs of doors from Experiment 3a, where both the *low-cost* (first column) and *no-cost* (second column) versions had the object in front of the door. Consistent with the main results, participants significantly inferred avoidance in the *low-cost* versions but not the *no-cost* versions. This suggests that having an object in front of a door is insufficient for triggering these inferences.

8. Experiment 3b: Replication with the Tsimane’

Experiment 3a suggests that people’s reasoning about low-cost communicative blockers cannot be reduced to explicit object-meaning conventions. However, it is possible that these inferences are culture-specific. Because our model is built on simple aspects of human cognition that are thought to be universal, the absence of these inferences in other cultures would challenge our account. As a first step in exploring this possibility, we replicated a variation of Experiment 3a with the Tsimane’—a farming-foraging group native to the Bolivian Amazon. The Tsimane’ live in non-industrialized communities along the Maniqui river and have less exposure to market-integrated communities compared to U.S. participants. Comparing the Tsimane’ and WEIRD participants (Western, educated, and from industrialized, rich, and democratic countries; Henrich et al., 2010) has helped identify cultural influences in color-word vocabulary (Conway et al., 2020; Gibson et al., 2017) and music perception (McDermott et al., 2016), and has also helped rule out cultural influences in other domains, such as the stages of number-word learning in children (Piantadosi et al., 2014; Jara-Ettinger et al., 2017) and the ways in which people identify communicative action (Royka et al., 2022). We therefore sought to test the Tsimane’ as a way to explore if these inferences also emerge in a culture that is substantially different from the U.S.

747 8.1. *Participants*

748 133 Tsimane’ adults were recruited in their local communities in the Bolivian
749 Amazon ($M = 33.12$ years, $SD = 15.40$ years). 17 additional participants were
750 recruited but excluded from the study for failing to complete the study (see
751 Results).

752 8.2. *Stimuli*

753 Stimuli consisted of six images of doors, each with an object in front of it
754 (e.g., Figures 7a-b). We used a subset of the objects used in Experiment 3a that
755 Tsimane’ participants were familiar with (as determined by our interpreters)
756 while remaining unconventional as communicative objects: a plant, a chair, and
757 a pile of cinderblocks. Each object was associated with two different doors:
758 a “low-cost door”, where the object was placed directly in the middle of the
759 doorway (e.g., Figure 7a, left), and a “no-cost door”, where the object was
760 placed next to the door, not blocking the way (e.g., Figure 7a, right). Half of
761 these doors were open and the other half were closed.

762 8.3. *Procedure*

763 The procedure was adapted from Experiment 3a to be more intuitive for our
764 participants, based on feedback from our interpreters. Participants were asked
765 to imagine deciding to enter a friend’s house through one of two possible doors.
766 Participants were then shown a low-cost door and a no-cost door sequentially
767 (order counterbalanced across participants, with different objects used for each
768 door) and were asked: “Do you think the owner wants you to enter or stay
769 away?”

770 8.4. *Results*

771 Participants that failed to complete both trials were excluded from the study
772 ($n = 17$; 11.33% exclusion rate; see SM for details on excluded participants).
773 Like U.S. participants, Tsimane’ participants inferred that they should avoid a
774 door when the object was minimally blocking the door (85.71%; $CI_{95\%}$: 79.70%–
775 91.73%; $p < 0.001$ from a two-tailed binomial test), but not when the object
776 was on the side of the door (30.08%; $CI_{95\%}$: 22.56% – 38.35%; $p < 0.001$ from
777 a two-tailed binomial test).

778 8.5. Discussion

779 These results suggest that, like U.S. participants, Tsimane’ participants also
780 inferred avoidance from objects that impose a low cost. Critically, this experi-
781 ment used objects that were familiar to the Tsimane’, but not typically used by
782 them to communicate (as determined by our local interpreters). This approach
783 followed the same logic as our design with U.S. participants, which also used
784 familiar objects but, critically, not ones typically used to communicate. This
785 enabled us to maximize *equivalence* (Matsumoto & Yoo, 2006; Van de Vijver
786 & Leung, 2021; Poortinga, 1989)—the goal of reaching similarity in conceptual
787 meaning across groups to support meaningful comparisons. At the same time, a
788 stronger test of our hypothesis would have included entirely novel objects, which
789 would have allowed us to test the nature of these inferences without any possi-
790 ble influence from prior object knowledge. These results, therefore, only provide
791 evidence that people can infer the communicative meaning of familiar objects
792 that are not typically communicative, and we do not know if these inferences
793 would extend to entirely novel objects.

794 9. Experiment 3c: Inferences from conventional communicative bloc- 795 kers

796 In Experiments 3a and 3b, we found evidence that people are sensitive to
797 the cost an object imposes when reasoning about its potential communicative
798 meaning. Importantly, these experiments used objects with no pre-existing
799 communicative meaning associated with them. Under our account, these infer-
800 ences become critical when people do not have a pre-existing convention, but
801 may become less important when they already know an object’s communicative
802 meaning. In Experiment 3c, we replicated Experiment 3a using conventional
803 objects. If people are constantly making cost-based inferences with all commu-
804 nicative objects, these results should replicate the pattern of Experiment 3a:
805 inferring avoidance in the *low-cost* condition, but not in the *no-cost* condition.
806 However, if these inferences are only at work when encountering novel objects,
807 people should report the conventional communicative meaning of the object
808 regardless of the cost that it imposes.

809 9.1. Participants

810 60 U.S. participants (as determined by their IP address) were recruited via
811 Prolific ($n = 80$ per condition; $M = 35.72$ years, $SD = 12.48$ years). 4 addi-

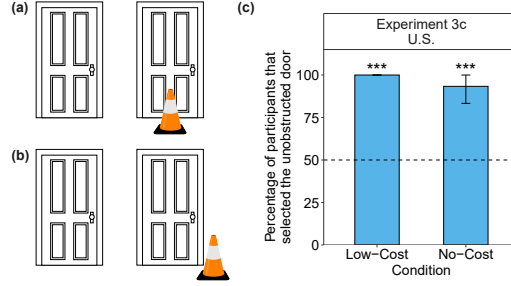


Figure 10: (a) Example stimuli from the *low-cost* condition in Experiment 3c. (b) Example stimuli from the *no-cost* condition in Experiment 3c. (c) Experiment 3c results. The blue bars represent the percentage of U.S. participants that selected the empty door as a function of condition. Error bars are bootstrapped 95% CIs (not visible in *low-cost* condition due to participants performing at ceiling).

812 tional participants were recruited and replaced for failing our inclusion criteria
813 (see Results).

814 9.2. Stimuli

815 Stimuli consisted of six images of pairs of doors, with each pair consisting
816 of an empty door and a door with an object nearby (similar to those used in
817 Experiment 3a, but using different objects; e.g., Figures 10a-b). Here we used
818 objects that are conventionally used as communicative blockers: a traffic cone,
819 construction tape, and a stanchion.

820 9.3. Procedure

821 The procedure was identical to that used in Experiment 3a, with the only
822 difference being the objects that participants saw. Participants then responded
823 to the same manipulation-check and inclusion questions from Experiment 3a.

824 9.4. Results

825 Participants who said the empty door was harder to walk through were ex-
826 cluded from the study and replaced ($n = 4$; 6.25% exclusion rate), since the
827 empty door is never harder to walk through (see SM for details on excluded
828 participants). Of our final sample, 100.00% of participants in the *low-cost* con-
829 dition inferred that they were supposed to avoid the door with an object ($CI_{95\%}$:

100.00% – 100.00%; $p < 0.001$ from a two-tailed binomial test². 93.33% of participants in the *no-cost* condition also inferred that the door should be avoided (CI_{95%}: 83.33% – 100.00%; $p < 0.001$ from a two-tailed binomial test. While the number of participants inferring that they should avoid the door was qualitatively higher in the *low-cost* condition relative to the *no-cost* condition, this difference was not significant ($p = 0.492$ by Fisher’s exact test). These results suggest that conventional knowledge was not driving our effect in Experiment 3a, and that people may have a lower reliance on costs when interpreting conventional communicative objects.

10. Experiment 4: Conventionalizing object meanings

Experiments 2, 3a, and 3b suggest that people are sensitive to costs when reasoning about an object with no pre-existing convention about its meaning. Experiment 3c further shows evidence that this sensitivity disappears when the object already has a conventional meaning associated with it. Experiment 4 presents an initial test of how objects that trigger inferences might become conventionalized (no longer requiring inference).

The process of associating objects with meaning might be particularly valuable, as it would help people minimize cognitive demands (Birch & Bloom, 2007; Back & Apperly, 2010; Keysar et al., 2000) and cognitive effort (Shenhav et al., 2017; Kool & Botvinick, 2018). That is, people might do social inference to interpret novel communicative objects, but treat them as conventional quickly afterwards. This hypothesis would help explain why some communicative objects, like chairs to indicate shoveled parking spots (e.g., Figure 1c), are used consistently (although not always; e.g., Figure 1i), and is parsimonious with a resource-rational view of the mind (Lieder & Griffiths, 2019; Griffiths et al., 2015).

Participants in Experiment 4 were first asked to infer the meaning of a low-cost object, and then completed a second trial that either showed a door with a picture of the same object (*congruent* condition) or a picture of a new object (*incongruent* condition). The critical idea in this experiment is that the picture in the second trial never imposes a cost. If participants interpret

²Our pre-registered analysis proposed to use *t*-tests to study this effect. We instead present binomial tests for clarity, but the results are identical under our pre-registered analysis and can be found in SM.

that picture independently, they should perform at chance when asked what it means (replicating the responses in the *no-cost* condition of Experiment 3a). Alternatively, people might infer the meaning of the low-cost object in the first trial and then immediately treat it as conventional. If so, then participants should report the same inference when they see a picture of the same object (i.e., in the *congruent* condition) despite the object not imposing a cost, but not when they see a picture of an unrelated object (i.e., in the *incongruent* condition).

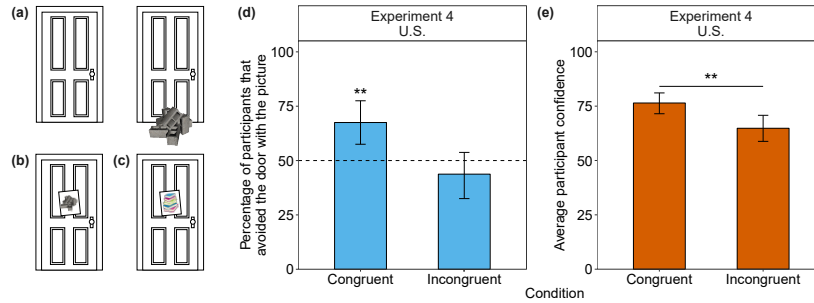


Figure 11: (a) Example stimuli from the first trial of Experiment 4. (b-c) Example of a door from the *congruent* and *incongruent* condition relative to (a), respectively. (d) Percentage of participants that inferred that they should avoid the door with the picture as a function of condition in Experiment 4. (e) Average reported confidence rating in this inference. Error bars are 95% bootstrapped CIs.

10.1. Participants

160 U.S. participants (as determined by their IP address) were recruited using Amazon Mechanical Turk ($n = 80$ per condition; $M = 36.13$ years, $SD = 10.90$ years). 42 participants were recruited and replaced for failing our inclusion criteria (see Results).

10.2. Stimuli

Stimuli consisted of 16 images of pairs of doors, with each object (using the same eight objects from Experiments 2 and 3a) associated with two pairs. In the “low-cost pair”, the object was placed directly in the middle of one of the doorways (e.g., Figure 11a), next to an empty door. In the “symbol pair”, a picture of the object was placed directly in the middle of one of the doorways (e.g., Figures 11b-c), next to an empty door. Half of these door pairs were open and the other half were closed (with the picture hanging from the top of the door frame when the door was open).

883 10.3. Procedure

884 The experiment began in an identical way to Experiment 3a, with the dif-
885 ference that the cover story named the person who had positioned the commu-
886 nicative object. This was done so that we could ensure participants understood
887 that the communicative objects came from the same agent across trials (name
888 randomized across participants).

889 In the first trial, participants saw a low-cost pair of doors (e.g., Figure 11a)
890 and were asked: “What do you think NAME was trying to tell you about the door
891 with the OBJECT?” (with the same possible responses from Experiment 3a) and
892 “How confident are you that that’s what NAME was trying to tell you?” (using
893 a continuous slider ranging from “not confident at all” to “very confident”).

894 Participants were told that their inference was correct (regardless of their
895 answer) and they were next presented with a symbol pair of doors. In the
896 *congruent* condition, one of the doors had a picture of the object from the
897 previous trial (e.g., Figure 11b after seeing Figure 11a, right, on the first trial).
898 In the *incongruent* condition, one of the doors had a picture of a new object (e.g.,
899 Figure 11c after seeing Figure 11a, right, on the first trial). Participants were
900 asked the same two questions from the first trial, followed by the manipulation-
901 check and inclusion questions from Experiments 2 and 3a.

902 10.4. Results

903 Participants who did not respond that the object in the first trial imposed
904 a cost were excluded from the study and replaced ($n = 42$; 20.79% exclusion
905 rate), as our interest is in how participants generalize the inferred meaning of
906 objects that they perceived as imposing a cost (see SM for details on excluded
907 participants).

908 The first trial replicated our results from Experiment 3a, with 66.25% ($CI_{95\%}$:
909 58.75% – 73.75%) of participants inferring that they should avoid the door with
910 the object ($p < 0.001$ from a two-tailed binomial test). Participants reported
911 an average confidence rating of 76.96% ($CI_{95\%}$: 73.71% – 80.03%).

912 We next turned to our main question of interest. If participants treat the pic-
913 ture in the second trial as a novel signal, they should perform at chance in both
914 conditions, as the picture does not impose a cost. By contrast, if participants as-
915 sume that the signaler was treating the object as a new convention, they should
916 infer that the door should be avoided in the *congruent* condition, but perform at
917 chance in the *incongruent* condition. As predicted, participants in the *congru-*
918 *ent* condition judged that the door with the picture should be avoided, despite

the picture not imposing a cost (67.50%; $CI_{95\%}$: 57.50% – 77.50%; $p < 0.01$ from a two-tailed binomial test). In the *incongruent* condition, only 43.75% of participants inferred that the door should be avoided ($CI_{95\%}$: 32.50% – 55.0%; $p = 0.314$ from a two-tailed binomial test). Moreover, participants were significantly more confident in their interpretation of the picture in the *congruent* condition (76.45%; $CI_{95\%}$: 71.46% – 81.09%) relative to the *incongruent* condition (64.83%; $CI_{95\%}$: 58.78% – 70.81%; $W = 3978$, $p < 0.01$ from a U -test).

11. Experiments 2-4 Discussion

Experiments 2-4 suggest that people can infer the potential communicative meaning of an object in the absence of explicit pedagogy and convention. People’s inferences were qualitatively consistent with our proposal, where communicative inferences are guided by reasoning about others’ mental states. Is it possible that participants arrived to these inferences through simpler heuristics?

A first concern is that an object placed in front of a door might be more salient than an object placed to the side. It is possible that participants in Experiment 2 thought that visually-salient objects (rather than low-cost objects) were more likely to be communicative. However, that it is unclear why people would expect visual salience to imply avoidance, and there are cases where visual salience is interpreted to mean the opposite (Misyak et al., 2016). In addition, a general expectation that communicative objects should be visually salient is not mutually exclusive with our account, and is in fact consistent with the main idea in our proposal: To elicit mental-state inferences through objects, the communicator must also ensure that the recipient will notice the object in the first place.

A second potential concern is that people find unusual arrangements of objects to be more likely to be communicative, and to signal avoidance. To test for this possibility, in Experiment 3a we collected “unusualness” ratings for our stimuli and found that the cost information significantly explained participant judgments when controlling for unusualness (see SM for details).

12. General Discussion

Human communication is remarkable in that, beyond words and gestures, we can communicate through objects. Here we proposed that this ability emerges

951 from our capacity to represent and infer other people’s mental states. Specif-
 952 ically, we proposed that, if we can reason about other people’s mental states
 953 based on how they manipulate objects in the environment, then people can also
 954 arrange objects with the purpose of revealing their mental states to agents who
 955 encounter these objects.

956 In exploring this idea, our paper makes three contributions. First, we imple-
 957 mented a computational model of social inference from physical objects. This
 958 model revealed how communicators can use objects to elicit mental-state infer-
 959 ences in observers, and how observers can infer the communicative meaning of
 960 objects, without any direct communication occurring between the agents. This
 961 provided proof-of-concept that the computations proposed in our account are
 962 sufficient to give rise to this phenomena. Second, we directly tested whether
 963 people’s inferences about the communicative meaning of an object could be ex-
 964 plained by our account. Finally, we tested whether people’s intuitions about
 965 communicative objects could only be the result of explicit systems of pedagogy
 966 and convention. Combined, these results suggest that mental-state inferences
 967 support the creation and interpretation of novel communicative objects, while
 968 pedagogy and convention drive their widespread use.

969 12.1. *Model assumptions and study limitations*

970 At its core, our model consisted of a Theory of Mind implementation which
 971 we built following several computational principles that each enjoy strong em-
 972 pirical support: social interactions involve agents thinking about each other’s
 973 mental states (Goodman & Frank, 2016; Frank & Goodman, 2012); mental-state
 974 reasoning is structured around an assumption that agents maximize utilities
 975 (Jara-Ettinger et al., 2016; Jern et al., 2017; Lucas et al., 2014); inferences about
 976 other people’s minds is performed via some approximation of Bayesian inference
 977 (Baker et al., 2009, 2017); and agents are typically cooperative, particularly
 978 when cooperation is easy (Powell, 2022; Rand, 2016; Warneken & Tomasello,
 979 2006). To capture how we might extract social information from the physical
 980 world, we made three further assumptions in our experiments and model.

981 Our first assumption was that agents can identify which objects in a scene
 982 were manipulated by an agent (as opposed to being the result of some inanimate
 983 force; e.g., the wind). This assumption was reflected in the formulation of our
 984 decider model, where deciders know both the initial scene s_0 and the final scene s
 985 (such that any discrepancy between s and s_0 reveals the costs that the enforcer
 986 incurred, and the costs that they introduced to the decider). While related

research shows that some form of this capacity emerges in infancy (Newman et al., 2010; Keil & Newman, 2015), it is likely that in more realistic contexts people do not know the initial scene s_0 and instead have a prior distribution over potential initial scenes $p(s_0)$.

Our second assumption was that the objects we considered were intentionally placed by an agent. In real-world situations, this is not known a priori and people must determine which objects were placed intentionally and which were not. Intuitively, there are many objects that, when encountered in front of a door, would not elicit a communicative inference because their placement appears unintentional. For instance, finding an empty paper bag (which an agent might have simply discarded), a soccer ball (which might have rolled over to the front of the door), or a wallet (which an agent might have dropped) would not trigger communicative inferences, because their placement does not seem intentional. Recent research has found that people see behavior as intentional when the outcome causally depends on the agent’s desires (Quillien & German, 2021) and when the agent’s behavior increases the odds of the outcome happening (Ericson et al., 2023). Integrating these types of processes into the detection of communicative objects is a key step towards having a more flexible framework that can both infer the meaning of communicative objects, and disregard objects that lack a communicative purpose.

Our third assumption was that people can estimate the cost associated with moving objects (for the enforcer) and navigating around them (for the decider), but our model did not explicitly capture how people determine these costs. Recent work shows that, from childhood, people might estimate the effort and difficulty of manipulating objects through an intuitive theory of physical action (Yildirim et al., 2019; Gweon et al., 2017), and integrating this work into our model may enable us to explain communicative objects in more complex situations. This is a direction we hope to pursue in future work.

Our work also has two important limitations. Our first limitation is that we focused our analysis on a specific class of objects where agents reveal their mental states by imposing minimal costs on observers, which we referred to as low-cost communicative blockers (LCCBs). These LCCBs are widespread and easy to find in our everyday lives (see Figure 1 for examples). Indeed, the presence of a minimal low-cost barrier can even reduce transgressions in children (e.g., children are less likely to peek at the answers in a desk if there is a minimal barrier placed between the two), although this effect even extends to imaginary barriers (Zhao et al., 2020).

1024 How might our framework extend to other types of communicative objects
1025 in broader contexts? Most directly, our framework can also explain cases where
1026 agents decrease a cost to signal invitation (e.g., leaving a box of cupcakes inten-
1027 tionally open to signal that anyone can grab one, or leaving an office door ajar
1028 to indicate we can be interrupted). From our model’s perspective, these infer-
1029 ences are symmetrical, with the only difference being that the observer would
1030 infer that a communicator intentionally lowered the cost (rather than increased
1031 it). This is a prediction that we hope to test in future work.

1032 The most general formulation of our proposal is not intrinsically tied to
1033 cost manipulation specifically, but rather to environmental manipulations that
1034 reveal mental states. Our work is thus consistent with related work showing
1035 that people can use no-cost markers to signal flexible meanings. For instance,
1036 when given a sticker to mark which of three cups someone should choose, people
1037 use the sticker to signal the right cup. By contrast, when one of the three cups
1038 must be avoided, people now use the sticker to indicate avoidance (Misyak
1039 et al., 2016). This has been hypothesized to reflect a process known as “virtual
1040 bargaining” (Misyak et al., 2016, 2014; Misyak & Chater, 2014), where people
1041 produce whichever solution they would have reached if they had gotten the
1042 opportunity to have an explicit discussion about it.

1043 More broadly, if people communicate through objects by attempting to elicit
1044 mental states in observers, then communicators must also pay attention to ad-
1045 ditional dimensions that we did not explicitly model. Specifically, for observers
1046 to infer mental states from an object’s position, the object must be noticeable,
1047 and appear intentionally placed. Otherwise, the object might be ignored or dis-
1048 missed. Intuitively, these features might also play an important role in the use
1049 of low-cost communicative blockers (Figure 1). These additional dimensions are
1050 vital for a full computational model of how we infer mental states from objects.

1051 A second key limitation is that our model represents objects in terms of
1052 the utilities they provide or impose on agents. This abstraction means that
1053 our model does not make any conceptual distinctions between different types of
1054 objects as long as they impose the same costs (or provide the same rewards). For
1055 instance, our current implementation represents a plastic cup and an expensive
1056 water bottle on a table as equivalent (because they impose the same physical
1057 cost), even though these two objects would likely elicit different mental-state
1058 inferences in observers (in one case, an observer might assume it is trash, while
1059 in the other it may be interpreted as a “spot saver”). Similarly, a delivery bag
1060 in front of a door may impose a small cost, but knowing what delivery bags are

would prevent us from inferring that this is a low-cost communicative blocker. While these examples demonstrate how representing an object’s category can strip it of a communicative purpose, there are also cases where representing an object’s category strengthens it. For instance, a sisal rope and red velvet rope can both be used to communicate avoidance, but the latter communicates it more strongly (and also carries the implication that there could be negative consequences if the message was ignored, possibly due to knowing that the object was costly and built specifically for this purpose). Intuitively, agents also strategically take this into account when deciding which objects to use to communicate. This is broadly consistent with our account, as it reveals further social reasoning about what observers might find easier to interpret, but is not yet captured by our model.

12.2. Open questions

Our work leaves several major questions open. A first open question stems from our focus on adults: What is the developmental trajectory of how people use communicative objects? Related research shows that the capacities necessary for these kinds of inferences emerge early in development. From early childhood, people can infer the presence of a hidden agent based on the structure of the environment (Newman et al., 2010; Keil & Newman, 2015; Saxe et al., 2005; Ma & Xu, 2013); we can estimate the difficulty associated with fulfilling different tasks (Gweon et al., 2017; Bennett-Pierre et al., 2018; Yildirim et al., 2019); and we can explain behavior in terms of unobservable mental states, like beliefs, desires, and intentions (Wellman, 2014; Gopnik et al., 1997). Most strikingly, recent work has shown that children can also use Theory of Mind to infer the transmission of ideas based on how different agents build similar artifacts (Pesowski et al., 2020; Schachner et al., 2018). This “intuitive archeology” likely shares a common basis with the inferences in our model, opening the possibility that even children can detect and infer the communicative meaning of objects through Theory of Mind. It may even be the case that the inferences we studied here are in fact an extension of people’s “intuitive archeology” and ability to reason about the history of objects.

A second open question is about the relative difficulty in creating versus interpreting communicative objects. While our computational model can explain both the creation and interpretation of communicative objects (as revealed in our model simulations), our experimental work focused exclusively on the second component. We did so because, under our account, all people ought to be

1097 able to infer the meaning of communicative objects, while the ability to generate
1098 them can be limited to a few individuals. We thus do not know the extent to
1099 which people can easily create ad-hoc communicative objects in new situations.
1100 It is possible that, when creating new communicative objects, people might
1101 err on the side of caution and prefer to make the costs higher than would be
1102 necessary, to minimize the chance of an agent not realizing the communicative
1103 content. As people become more confident that the objects they place are being
1104 recognized as communicative, they may subsequently lower the costs. We hope
1105 to explore these questions in future work.

1106 A third open question is whether the inferences that we studied here are
1107 an extension of pragmatics in language. People’s ability to derive non-literal
1108 meaning in language is supported by a form of recursive reasoning, captured
1109 by the Rational Speech Act (RSA) framework (Franke & Jäger, 2016; Scontras
1110 et al., 2018; Frank & Goodman, 2012; Goodman & Frank, 2016). Our model can
1111 be thought of as an RSA model where the medium people use to communicate
1112 is objects (instead of utterances) and our costs are physical effort (rather than
1113 memory retrieval or utterance length). At the same time, the ability to perform
1114 recursive social inference is not unique to language and it is possible that the
1115 phenomena we studied here reflect a more general, non-linguistic Theory of
1116 Mind. As such, these phenomena might constitute a more primitive form of
1117 communication that precedes the ability to do pragmatics in language (i.e., a
1118 kind of proto-communication). This is a question that goes beyond the scope
1119 of our work.

1120 Finally, our work leaves open the question of how to capture other types of
1121 inferences that people make from objects. Intuitively, communicative inferences
1122 are only a sliver of the social information that we can read from objects. For
1123 instance, objects can also lead us to infer aesthetic goals (e.g., placing an object
1124 that we like in a visible location within our house), functional goals (e.g., leav-
1125 ing objects like winter gloves next to our front door for convenience), or even
1126 personality traits (e.g., inferring that someone is messy based on the general
1127 pattern of objects on their desk; Gosling et al., 2002). It is possible that these
1128 inferences could be captured through a richer model of Theory of Mind that
1129 can consider a broader set of goals that agents have when interacting with the
1130 environment, and this is a question that we hope to pursue in future work.

1131 13. Concluding Remarks

1132 Humans have a remarkable capacity to share their mental states through
1133 their behavior, language, and even the way they arrange objects in their en-
1134 vironment. Our work shows one way in which people can share their mental
1135 states through objects. And yet, the types of meaning that we give objects
1136 is even broader than what we show here—a metal band can signify a lifelong
1137 vow, a chiselled stone can commemorate a lost loved one, and a menorah can
1138 reveal one’s metaphysical beliefs. We hope that our work is a step towards
1139 understanding the rich social nature of the physical world.

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